PROJECT PROPOSAL

FLOODSHADOW

Rainfall awareness and territorial priorities for Guayaquil

One connected workflow to anticipate rainfall, follow changing conditions and identify places that deserve a closer look.



Guayaquil waterfront · Context photograph by Freddy eduardo · CC BY-SA 4.0 · Cropped for layout.

BEFORE
Anticipate

→

DURING
Follow

→

AFTER
Prioritize

TEAM 3JOTA · ECUADOR CHALLENGE

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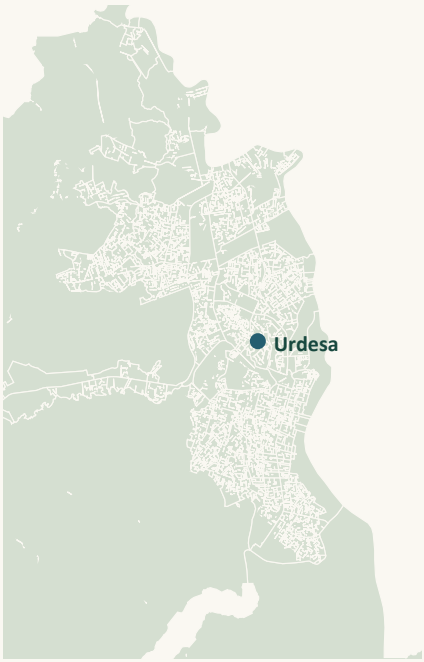
Hack2Impact 2026 · Project submission · 26 October 2026

01 PROBLEM AND SOLUTION

A clearer picture of intense rainfall

Guayaquil needs to connect what the weather may bring with what is happening on the ground and what should be checked next.

A forecast, a city map and a collection of incident reports describe different parts of the same storm. When those pieces remain separate, a place with few reports can be overlooked. FLOODSHADOW brings them into one geographic workspace, using recognizable areas and a clear sequence of decisions.



PRIMARY USERS

Local analysts, community coordinators and teams responsible for reviewing rainfall-related conditions. The focus is Guayaquil: a selected area can be followed across the three stages without switching to a different interface.

01	02	03
Before	**During**	**After**
What rain is expected?	What is changing now?	Where should we look next?

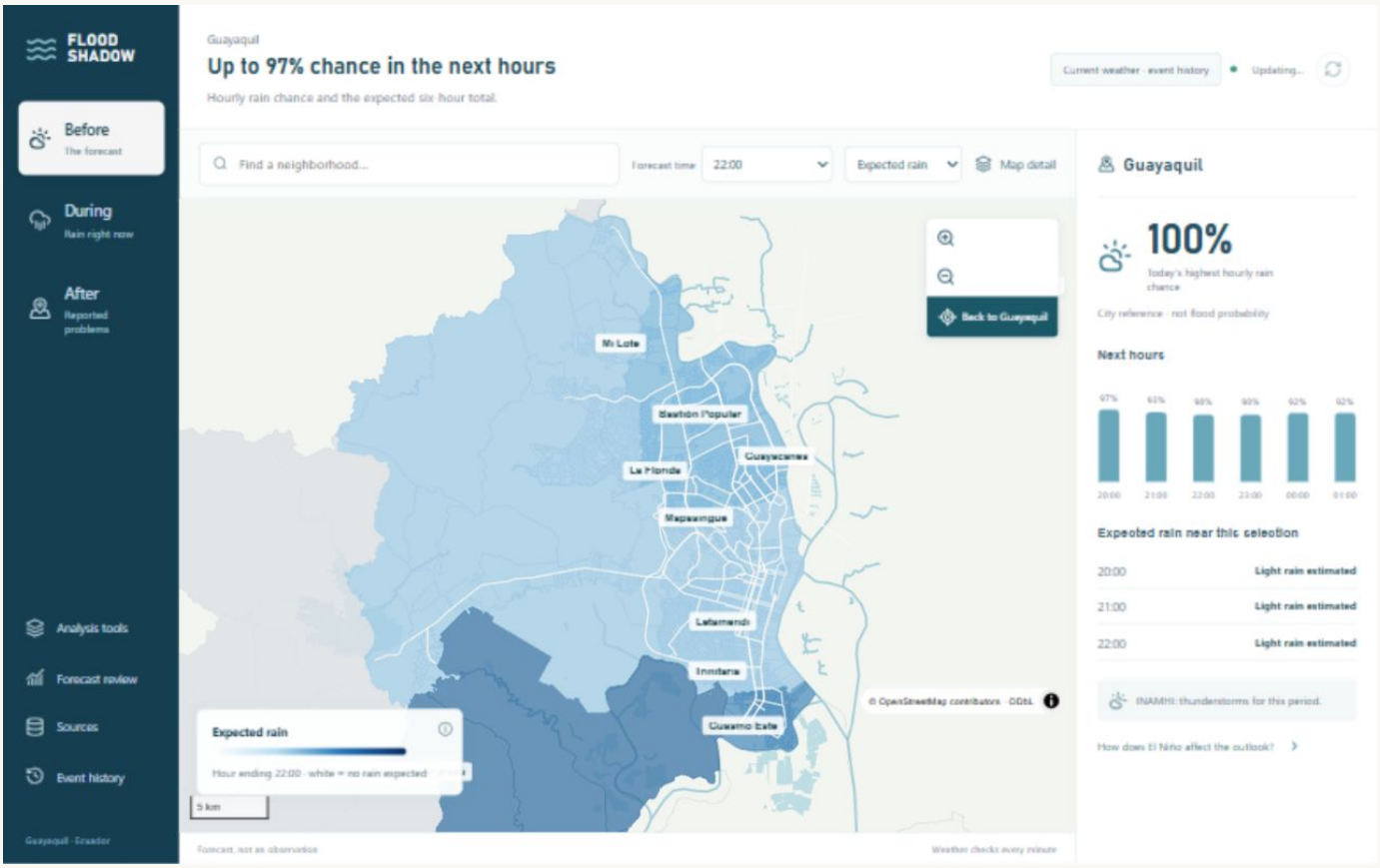
The proposed innovation is **attention to possible blind spots**: combining rainfall and territorial exposure with the availability of evidence, so the places needing verification can be visible even when published reports are sparse.

Reading route · Operating flow 03–05 · Territorial reasoning 06 · Use case 07 · Impact and plan 08 · Models and tools 09 · Evidence 10 · Sources 11

02 BEFORE

Anticipate the next six hours

Before turns a weather outlook into a time-based view of the city: when rain is expected, how much is forecast and which areas deserve attention.



Application capture · 26 October 2026 · The displayed conditions belong to the captured source times.

Read the outlook by hour

The default map combines the next six available hourly forecasts. Selecting an individual hour changes the displayed rainfall for that same time. The probability bars help the reader locate the hours when rain is more likely.

Connect weather with place

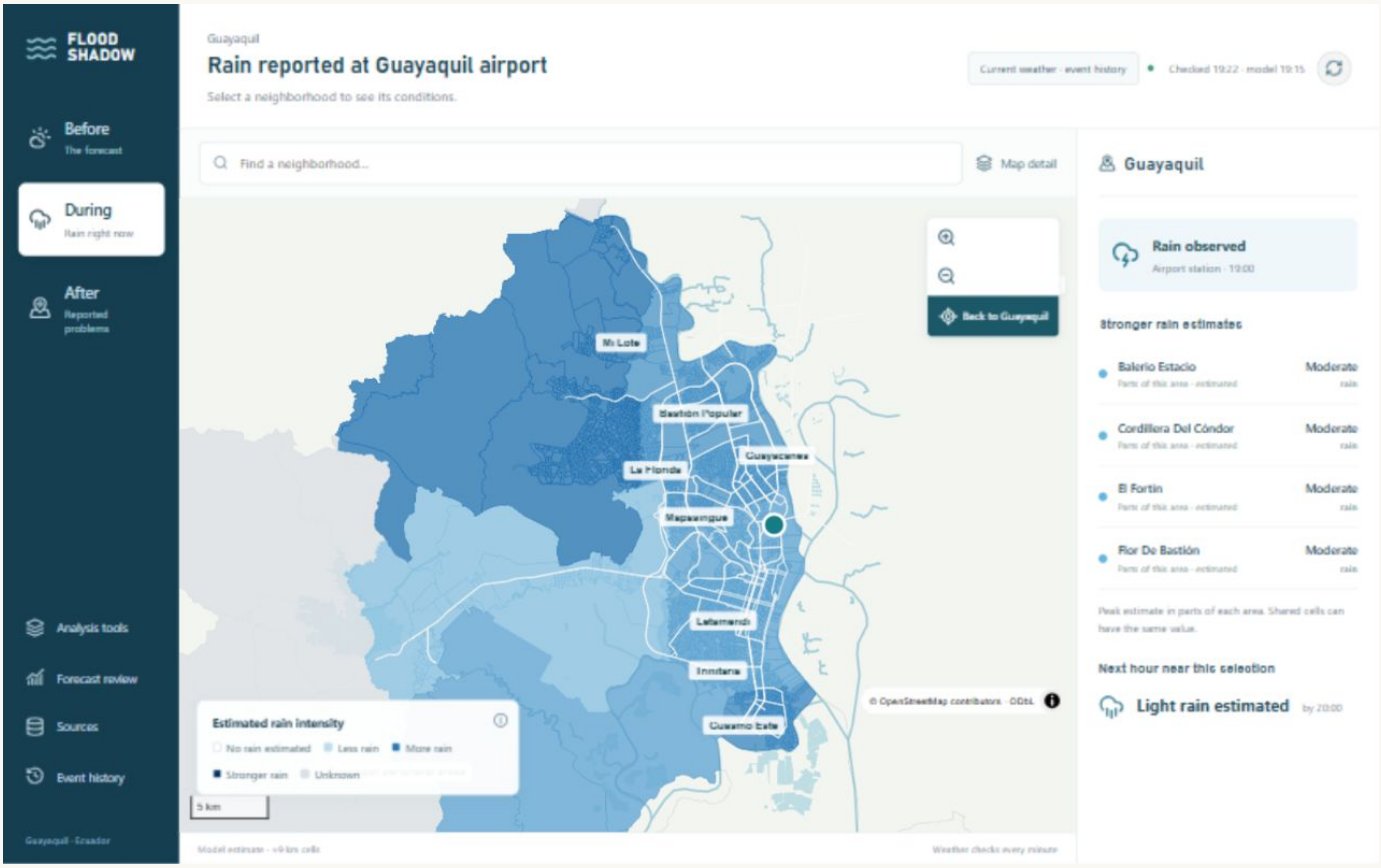
Search for an area to inspect its expected conditions. The optional watch view combines forecast rainfall with mapped exposure, making it easier to decide where to keep attention before the event develops.

+1 h · +2 h · +3 h · +4 h · +5 h · +6 h

03 DURING

Follow the rainfall as conditions evolve

During keeps the same city map and changes the question: which areas show stronger estimated rain at the latest available source time?



Application capture · 26 October 2026 · The displayed conditions belong to the captured source times.

A map that updates

The open view checks its weather connection every minute. When new source values are available, sector colors and the area summary update. The time of the source remains visible alongside the time of the latest check.

Two complementary views

Spatial model estimates show how rainfall varies across the city. The airport weather report provides an independent point observation. Selecting a named area focuses its polygons and makes its current conditions easier to inspect.

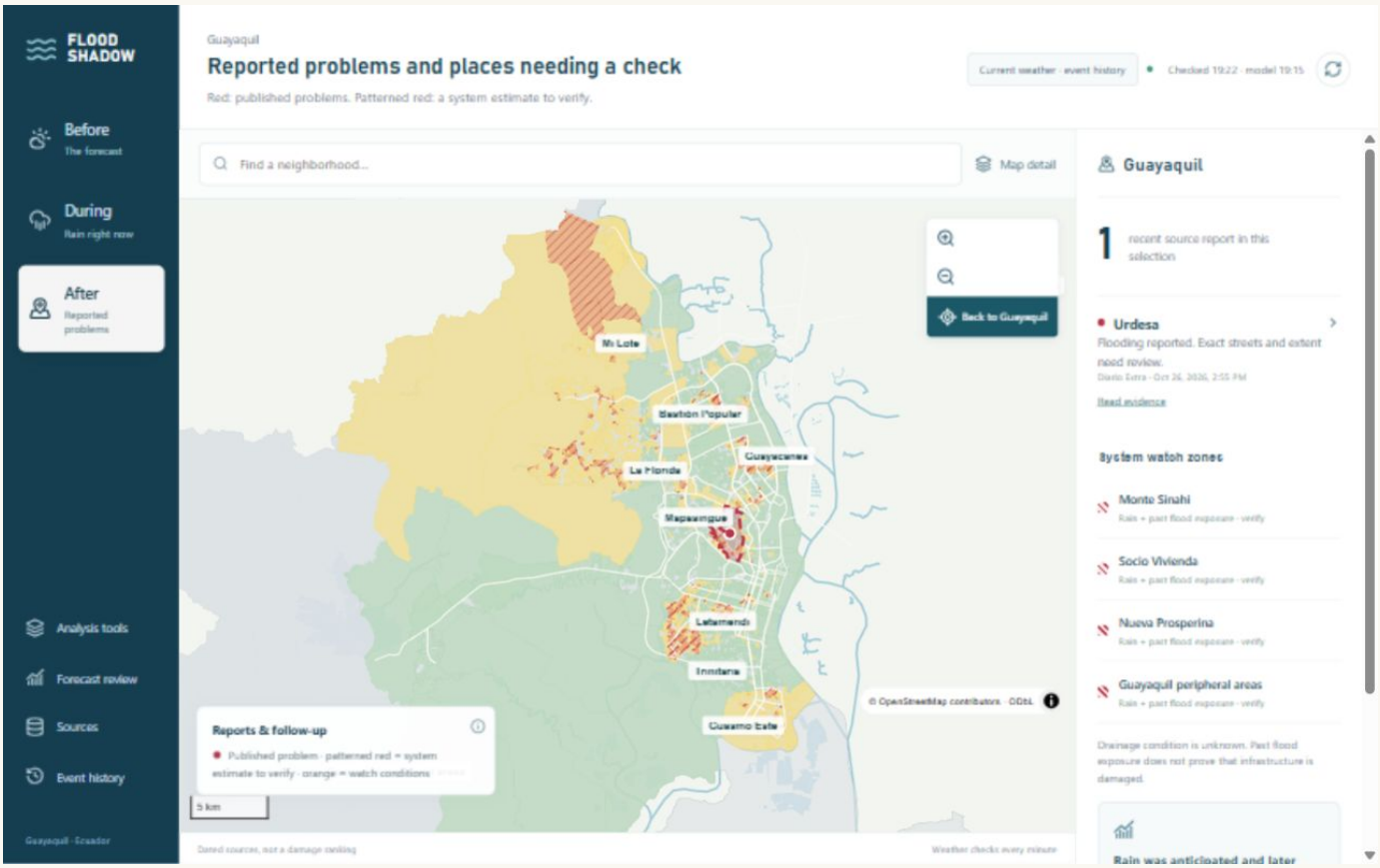
HOW TO READ THE BLUES

- **Light**
- **Moderate**
- **Heavy**

04 AFTER

Turn the event into follow-up priorities

After connects rainfall, exposed territory and published evidence to show where a reported problem exists and where verification deserves priority.



Application capture · 26 October 2026 · The displayed conditions belong to the captured source times.

Published evidence

Solid red refers to an area mentioned in a published problem report. A source can describe an incident inside an area without establishing that every street in that area was affected.

System watch zones

Patterned red points to specific polygons suggested for verification where fresh estimated rainfall overlaps mapped exposure. It directs the next check; it is not a measured damage total.

Solid red

Published problem

Patterned red

Priority for verification

05 TERRITORIAL REASONING

How the system builds a watch priority

FLOODSHADOW relates the expected or recent rain to geographic exposure, then keeps that reasoning connected to the evidence available for the selected event.

01 **Weather** — Forecast or recent modeled rainfall provides the weather context for each covered location.

02 **Territory** — Official census polygons and recognizable local areas organize the analysis geographically.

03 **Exposure** — A mapped susceptibility footprint identifies where rainfall may deserve closer attention.

04 **Evidence** — Published reports and reviewed checks distinguish documented problems from places still needing verification.

A USEFUL OUTPUT

The result is an **explainable order of attention**. For a flagged place, the reader can inspect the rain context, the territorial exposure and the available evidence rather than relying on a color alone.

This priority can guide practical checks of drainage, street access, housing and nearby essential services. Infrastructure condition is established through that review; the on-screen watch priority is derived from rainfall and mapped exposure.

06 A CONNECTED USE CASE

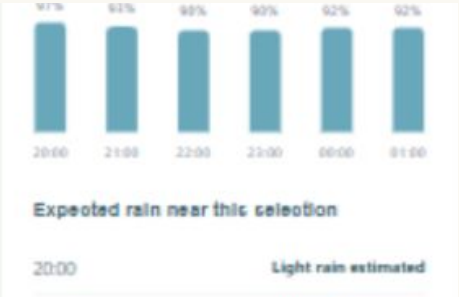
Follow one area through the same storm

The workflow keeps place, time and evidence together. Urdesa illustrates how a user can move from an outlook to current conditions and then to the follow-up view.

BEFORE

Choose the area and hour

Search for Urdesa. Read the expected rainfall and the next six hourly probabilities. Keep the selected hour in view when comparing the map.



DURING

Follow the source update

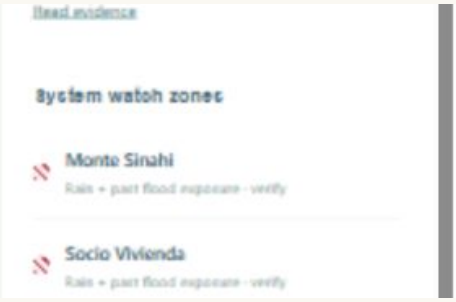
Open During without changing the geographic focus. Inspect the latest supported rainfall estimate and its source time. Use the airport report as a separate observation.



AFTER

Review evidence and the next check

Open After. Read the published Urdesa reference and inspect nearby system watch zones. Follow the source link and identify what needs a localized check.



The Urdesa report provides a published reference for the event. System watch zones add a territorial priority derived from rainfall and mapped exposure. Each highlight can be followed back to its weather context, geographic area and available evidence.

The operational outcome is a clear next action: inspect the area-specific evidence, record the check and retain the event context for later review.

07 EXPECTED IMPACT AND IMPLEMENTATION

From a clearer map to a better next decision

The intended benefit is faster, more traceable interpretation of a rainfall event and a stronger basis for selecting places to verify.

For local analysis

Reduce the effort needed to connect a forecast, a place and a source. Keep reported problems and information gaps in the same view.

For field coordination

Direct attention to exposed places that may otherwise receive few reports. Share a map and a reason for each proposed check.

IMPLEMENTATION PATH

1. **Operate the connected workflow**

   Maintain the weather connection, source dates and one-map navigation. Preserve event snapshots and source links.

2. **Review with local users**

   Test whether users correctly read the stages, the color legend and the difference between estimated and reported conditions.

3. **Verify selected places**

   Collect localized positive and negative checks, with place, time, scope and evidence. Retain corrections without erasing earlier records.

4. **Evaluate and extend**

   Compare saved outlooks with independent later evidence. Add verified infrastructure or other local data when coverage and reuse terms are established.

Success measures: time to find the relevant source, correct interpretation of the map, completion of representative checks and useful discoveries within a fixed verification effort. These are evaluation targets, not already measured emergency outcomes.

08 MODELS AND TOOLS

Public data connected to a readable city map

The implementation combines an external weather model with territorial reasoning. The interface explains their outputs through the same three-stage workflow.

Weather model

ECMWF IFS data through Open-Meteo supply hourly forecasts and spatial rainfall estimates. NOAA airport reports add an independent point observation; INAMHI guidance provides local meteorological context.

Territorial model

A transparent watch index combines rainfall and mapped exposure. Source-linked findings and verification states keep the outcome inspectable. It is a prioritization method, not a calibrated street-level flood probability.

INFORMATION FLOW

1 → 2 → 3 → 4

Collect
Public sources

Align
Place and time

Relate
Rain and exposure

Display
Map and evidence

CORE TECHNOLOGIES

React and Vite organize the web interface. **MapLibre GL JS** renders the interactive map. **Node.js** handles source refreshes. **Python geographic tools and SQLite** prepare and organize territorial information and saved records.

DEVELOPMENT TOOLS DECLARED

OpenAI Codex assisted coding and documentation. Team 3Jota is responsible for the project decisions and for explaining the implementation. Generative assistance is not used as a source of rainfall observations or incident evidence.

09 EVIDENCE AND DESIGN DECISIONS

A demonstration the team can explain

The project can be demonstrated through real interface states, source timestamps, named-area selection and preserved event records.

What can the team demonstrate?

Switch Before, During and After within one workspace; choose a forecast hour; focus a named area; inspect sources; and return to the city-wide map. The captures in this proposal show these connected views.

What does an automatic update mean?

The open application checks the weather connection every minute. A new check can return the same source value, so the interface keeps retrieval time and source time distinct.

Why distinguish solid and patterned red?

A published problem and a system suggestion carry different evidence. The legend keeps both useful without presenting a verification priority as confirmed damage.

How is an earlier forecast reviewed?

Saved forecasts retain their original values and receipt times. The review compares compatible later evidence with that saved outlook rather than rewriting what was predicted.

What makes the approach transferable?

The same workflow can be applied to another city once geography, sources and verification practices are adapted. A shared interface does not replace local data review.

Implementation checks cover missing and stale values, forecast-hour changes, source failures, event preservation and geographic filtering. Functional checks establish software behavior; predictive performance is evaluated separately against independent event evidence.

10 SOURCES AND CREDITS

Where the information comes from

Sources are linked to their role in the project. Each event retains its dates and evidence so the map can be read in context.

Open-Meteo
Hourly outlooks and ECMWF model-based rainfall estimates.
open-meteo.com/en/docs

NOAA Aviation Weather Center
Airport weather observations and METAR history for SEGU.
aviationweather.gov/data/api

INAMHI
Daily meteorological guidance and Ecuador forecast context.
servicios.inamhi.gob.ec/pronostico

INEC Geoportal
Official 2022 census-sector geometry used to organize the territory.
INEC · Marco Geoestadístico 2022 · Sector censal

Municipality of Guayaquil
Local sector nomenclature and recognizable geographic names.
sil.guayaquil.gob.ec · Sectores de nomenclatura

Zonas Inundables GYE geographic layer
Published susceptibility footprint used as territorial context.
ArcGIS · Zonas_Inundables_GYE

Diario Extra
Source-reviewed Urdesa reference in the recent-event view.
extra.ec · Guayaquil storm report · 26 October 2026

OpenStreetMap and MapLibre
Map attribution and the open-source geographic display.
maplibre.org

Event reference: Hack2Impact 2026 rules supplied by the team. Sections 4.1 and 5 define the Ecuador challenge and proposal contents; section 7.3 defines the judging criteria. This proposal covers the project document; the required English video pitch is a separate deliverable.

Visual credits: Application screenshots and diagrams: FLOODSHADOW / 3Jota. Basemap: OpenStreetMap contributors. Cover photograph: Freddy eduardo, *Vista del Malecón 2000*, Wikimedia Commons, 2 February 2024. Cropped photograph under CC BY-SA 4.0. Original image and author.